Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10212CS292 / Applied Programming Skills

Unit IV: Network and Relationship-Driven Data Structures
Course Outcome: Utilize graph and set-based structures to analyze connectivity, traversal, and relationship-driven problems

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 7	Session 31	Graph	Task 54: **Find if Path Exists in Graph** Task 55: **Find the Town Judge**	Leet Code
	Session 32	Breadth First Search	Task 56: **Rotting Oranges**	
	Session 33	Depth First Search	Task 57: **Number of Provinces** Task 58: **Max Area of Island**	
	Session 34	Connected components	Task 59: **Keys and Rooms**	
	Session 35	Cycle detection	Task 60: **Course Schedule** Task 61: **Course Schedule II**	

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 8	Session 36	Topological sorting	Task 62: **Sort Items by Groups Respecting Dependencies**	takeuforward / Leet Code
	Session 37	Shortest path in unweighted graphs	Task 63: **Shortest Path with Alternating Colors** Task 64: **01 Matrix**	
	Session 38	Grid traversal (islands, flood fill)	Task 65: **Number of Islands** Task 66: **Flood Fill**	
	Session 39	Union Find (Disjoint Set Union)	Task 67: **Accounts Merge**	
	Session 40	HashSet patterns – Frequency map and sorting by keys/values	Task 68: **Happy Number** Task 69: **Unique Email Addresses** Task 70: **Group Anagrams**	